

Dynamics and Control of a Nuclear Refueling Machine

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Abstract— This paper presents an active control scheme for suppressing the vortex-induced vibrations of a nuclear fuel rod transported under water. A refueling machine (RM), which is an overhead crane, is used for transporting the rod, vertically, such that one end of the rod is fixed to the RM and the other one hangs free (i.e., in cantilever configuration). First, Hamilton's principle is used in developing a hybrid lumped-mass (i.e., the RM) and distributed parameter (i.e., the rod) model of the considered system. Then, a boundary control scheme is developed to suppress the vortex-induced vibrations of the rod by utilizing control actuation only at the top boundary (i.e., the rod/RM interface). Further, Lyapunov function-based analyses are performed to prove the uniform ultimate boundedness of the closed-loop system. To demonstrate the effectiveness of the proposed control scheme, simulations have been performed.

I. INTRODUCTION

Whenever an object is moved in water, the fluid-structure interaction results in the generation of hydrodynamic forces, for example, the added mass force, the viscous damping force, the drag force, the buoyancy force, the vortex-induced force etc. Such forces not only oppose the motion but also result in the vibrations, both transient and residual, of the object. This paper is concerned with the vibration control of fuel rods, during a refueling process in a nuclear reactor, which are transported under water to minimize radiation leakage. Fig. 1 depicts a refueling machine (RM), a type of overhead cranes, transporting a master fuel assembly (MFA) in water to the desired locations within a reactor. For such refueling operations, vibration suppression of the fuel rod is crucial in terms of both preventing structural damages to the rod and minimizing operational delays in the refueling process. Although residual vibrations can be suppressed by implementing open-loop control methods; however, such control methods are not capable of suppressing transient vibrations like vortex-induced vibrations (VIVs). Therefore, this paper presents an active vibration control scheme for the fuel rods, in the presence of nonlinear hydrodynamic forces, by controlling the movements of the bridge and the trolley of the RM.

The vortex-induced force is generated perpendicular to the direction of motion of the rod, which is a consequence of shedding of vortices from the structure of the rod. The vortex-shedding phenomenon results in the formation of an

unstable wake region around the rod, where the wakes are generated at different frequencies and amplitudes resulting in different vortex patterns [1]. When the vortex-shedding frequency reaches any vibration mode of the structure, exaggerated VIVs of the rod are caused (i.e., the lock-in phenomenon). Therefore, an appropriate control strategy must be developed to suppress the VIVs of the rod during its underwater transport.

Several control techniques, both passive [2-3] and active [4-7], have been reported in the literature for circular cylinders/rods undergoing VIVs. However, for the considered RM system, passive methods, which require mounting of damping devices like spoilers and strakes along the length of the fuel rod, are not feasible. Therefore, active suppression of the VIVs of the rod is pursued in this research. In [4], an active velocity feedback control strategy was reported for a flexible cylinder undergoing VIVs in a wind tunnel. The said control scheme could successfully attenuate the vibration of a single vibration mode by enhancing the overall viscous damping characteristics of the considered system. Later, in [5], another active velocity feedback control scheme was developed to suppress multiple (dominant) vibration modes of a flexible cylinder by utilizing multiple piezoelectric sensors and actuators. Later, a more robust continuous sliding mode control method was developed in [6] for suppressing VIVs caused by immeasurable flow-induced disturbances. In [7], a robust adaptive algorithm was developed for rejecting the periodic vortex-induced excitations of a flexible cylinder, considering parametric uncertainties.

The control schemes discussed thus far were developed to suppress the VIVs of a flexible cylinder in air. According to [8], a flexible body exhibits larger deflections in water as compared to those in air upon an excitation of same magnitude. Therefore, considering underwater operation of the RM system, some important works addressing suppression of VIVs in water are thereby discussed: In [9], an adaptive control scheme was developed to stabilize a flexible cylinder under hydrodynamic loads by manipulating the position of a tuned mass damped attached to the structure. In [10], a linear quadratic regulator (LQR) based control law was developed to stabilize a riser under sea loads. In [11], an active control strategy was proposed to suppress the resonant mode of a marine riser undergoing VIVs by displacing the top end of the riser. Other than the above mentioned control schemes, adaptive fuzzy sliding mode [12] and time-delay feedback [13] control schemes have also been recently proposed to actively mitigate the VIVs of flexible structures in water.

This work was supported by the National Research Foundation (NRF) of Korea under the auspices of the Ministry of Science and ICT, Republic of Korea (grant nos. NRF-2017R1A2A1A17069430 and NRF-2017R1A4A1015627)

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Most of the works, discussed above, have considered only the dynamics of the cylinder, given by partial differential equations (PDEs), in formulating the control model. However, in this paper, the control model of the RM system also includes the dynamics of the trolley and bridge of the RM, represented by ordinary differential equations (ODEs). Hamilton's principle is used to obtain a hybrid ODE-PDE model of the system to depict the deflections/vibrations of the flexible rod, which are coupled with both the movements of the RM and the hydrodynamic forces. To suppress the vibrations of the rod, a boundary control scheme is developed in this paper, which suppresses the VIVs of the rod by utilizing the movements of the RM as the rod is transported to the desired underwater location within the reactor. It is shown, by Lyapunov function-based analyses, that the closed loop system is uniformly ultimately bounded in the presence of nonlinear hydrodynamic forces. The validation of the proposed control scheme is demonstrated by performing simulations.

II. PROBLEM FORMULATION

Fig. 1 depicts the schematic of the RM, where the MFA is assumed to be an Euler-Bernoulli beam with a circular cross-section. Let l , d , I , E , and m_r denote the length, the diameter, the area moment of inertia, the Young's modulus, and the mass of the rod, respectively; $u(x, t)$ and $w(x, t)$ imply the lateral and transverse deflections of the rod; $y(t)$ and $z(t)$ indicate the displacements of the bridge and the trolley moving along the j and k axes; m_b and m_t be the masses of the bridge and trolley, and $f_y(t)$ and $f_z(t)$ denote the associated control inputs, respectively. In the following derivation of the equations of motion of the RM system, the derivatives with respect to t and x are denoted by $\dot{}$ and $\prime$, respectively, for simplicity.

The kinetic energy (T) of the bridge, the trolley, and the rod is given as follows.

$$T(t) = (1/2)((m_b + m_t)\dot{y}(t)^2 + m_t\dot{z}(t)^2 + m \int_0^l ((\dot{y}(t) + \dot{u}(x, t))^2 + (\dot{z}(t) + \dot{w}(x, t))^2) dx), \quad (1)$$

where m is the (combined) mass per unit length of the rod and the fluid displaced by the rod. The potential energy (U) of the rod is obtained as

$$U(t) = (EI/2) \int_0^l (u''(x, t)^2 + w''(x, t)^2) dx. \quad (2)$$

Assumption 1 [14]. If the kinetic energy T of the system in (1) is bounded for $\forall t \in [0, \infty)$, then $\dot{u}(x, t)$ and $\dot{w}(x, t)$ are bounded for $\forall t \in [0, \infty)$.

Assumption 2 [14]. If the potential energy U in (2) is bounded for $\forall t \in [0, \infty)$, then $u''(x, t)$, $u'''(x, t)$, $w''(x, t)$, and $w'''(x, t)$ are bounded for $\forall t \in [0, \infty)$.

Let c be the viscous damping coefficient in water. Then, the damping forces in opposition to the motion of the rod along lateral and transverse directions are given as $-c\dot{u}(x, t)$

and $-c\dot{w}(x, t)$, respectively [15]. According to Hamilton's principle, the following equations of motion are obtained.

$$M_1\ddot{y}(t) + m \int_0^l \ddot{u}(x, t) dx = f_y(t), \quad (3)$$

$$M_2\ddot{z}(t) + m \int_0^l \ddot{w}(x, t) dx = f_z(t), \quad (4)$$

$$EIu''''(x, t) + c\dot{u}(x, t) + m\ddot{u}(x, t) = -m\ddot{y}(t) - f_{h,y}(x, t), \quad (5)$$

$$EIw''''(x, t) + c\dot{w}(x, t) + m\ddot{w}(x, t) = -m\ddot{z}(t) - f_{h,z}(x, t). \quad (6)$$

Equations (3)-(4) represent the dynamics of the bridge and the trolley, whereas (5)-(6) depict the dynamics of the rod along the j - and k -axis, respectively; where $M_1 = m_b + m_t + ml$, $M_2 = m_t + ml$, and $f_{h,y}$ and $f_{h,z}$ are the hydrodynamic forces acting on the rod along the j - and k -axis, respectively.

When the rod is moved in water, the fluid-structure interaction results in generation of hydrodynamic forces both along (i.e., inline force f_I) and perpendicular (i.e., transverse f_T) to the direction of motion of the rod. The inline force is given by the Morison equation as [18]

$$f_I(x, t) = (\pi/4)\rho_w C_a(x) d^2 \dot{v}(x, t) + (1/2)\rho_w C_d(x) d v(x, t) |v(x, t)|, \quad (7)$$

where ρ_w is the density of water, C_a is the added-mass coefficient, C_d is the drag coefficient, and $v(x, t)$ is the velocity of the rod. The transverse force is a result of shedding of vortices from the structure of the rod, which is also termed as the vortex-shedding or lift force in the literature, and is given as follows [16].

$$f_T(x, t) = (1/2)\rho_w C_L(t) d v^2(x, t) \cos(\omega_v t + \phi), \quad (8)$$

where $C_L(t)$ is the time-varying lift coefficient, ϕ is the phase angle, and ω_v is the frequency of vortex shedding given by.

$$\omega_v = 2\pi St v(x, t) / d, \quad (9)$$

where St is the Strouhal number. In (8), the lift coefficient is given as follows.

$$C_L(t) = a_1 \cos(\omega_v t) + a_2 \cos(\omega_v t + \pi/2), \quad (10)$$

where $a_1 = 1.23$ and $a_2 = 0.042$ [17].

The considered system, (3)-(6), is highly coupled: In air, when there are no hydrodynamic forces, the movements of the RM result only in the deflections of the rod. However, in water, both the vibrations of the rod and the generation of hydrodynamic forces are associated with movements of the RM. For instance, a movement of the bridge alone will generate the inline force along the j -axis (i.e., $f_{I,y}$) and the transverse force along the k -axis (i.e., $f_{T,z}$), which are given below.

$$f_{I,y}(x, t) = (\pi/4)\rho_w C_a(x) d^2 (\ddot{y}(t) + \ddot{u}(x, t)) + (1/2)\rho_w C_d(x) d (\dot{y}(t) + \dot{u}(x, t)) |\dot{y}(t) + \dot{u}(x, t)|, \quad (11)$$

$$f_{T,z}(x, t) = (1/2)\rho_w C_L(t) d (\dot{y}(t) + \dot{u}(x, t))^2 \cos(\omega_v t + \phi). \quad (12)$$

Similarly, a movement of only the trolley will generate the inline force along the k -axis (i.e., $f_{I,z}$) and the transverse force along the j -axis (i.e., $f_{T,y}$) [18]:

$$f_{I,z}(x,t) = (\pi/4)\rho_w C_d(x)d^2(\ddot{z}(t) + \ddot{w}(x,t)) \\ + (1/2)\rho_w C_d(x)d(\dot{z}(t) + \dot{w}(x,t))|\dot{z}(t) + \dot{w}(x,t)|, \quad (13)$$

$$f_{T,y}(x,t) = (1/2)\rho_w C_L(t)d(\dot{z}(t) + \dot{w}(x,t))^2 \cos(\omega_v t + \phi). \quad (14)$$

The inline force consists of two parts: The inertial force (f_M) and the drag force (f_D), given by the first and the second terms in (7). Therefore, (11) and (14) can be re-written as $f_{I,y}(x,t) = f_{M,y}(x,t) + f_{D,y}(x,t)$ and $f_{I,z}(x,t) = f_{M,z}(x,t) + f_{D,z}(x,t)$, respectively. Since the inertial force of the fluid is already incorporated in m (i.e., $m = (m_r/l) + (\pi/4)\rho_w C_d(x)d^2$), the hydrodynamic forces along the j - and k -axis can be written as $f_{h,y}(x,t) = f_{D,y}(x,t) + f_{T,y}(x,t)$ and $f_{h,z}(x,t) = f_{D,z}(x,t) + f_{T,z}(x,t)$, respectively.

Assumption 3. There exist positive constants $\bar{f}_{h,y}, \bar{f}_{h,z} \in R^+$ such that $\|f_{h,y}(x,t)\| \leq \bar{f}_{h,y}$ and $\|f_{h,z}(x,t)\| \leq \bar{f}_{h,z}$, $\forall (x,t) \in [0,l] \times [0,\infty)$.

Substituting $\ddot{y}(t)$ of (3) into (5) results in the following equation representing the coupled lateral dynamics of the RM, the rod, and the surrounding water.

$$EIu''''(x,t) + cu(x,t) + m\ddot{u}(x,t) = -f_{h,y}(x,t) \\ - (m/M_1)(f_y(t) - m \int_0^l \ddot{u}(x,t) dx). \quad (15)$$

Similarly, substituting $\ddot{z}(t)$ of (4) into (6) results in the following equation representing the coupled transverse dynamics of the considered system.

$$Elw''''(x,t) + c\dot{w}(x,t) + m\ddot{w}(x,t) = -f_{h,z}(x,t) \\ - (m/M_2)(f_z(t) - m \int_0^l \ddot{w}(x,t) dx). \quad (16)$$

The boundary conditions are obtained as follows.

$$u(0,t) = u'(0,t) = u''(l,t) = u'''(l,t) = \\ w(0,t) = w'(0,t) = w''(l,t) = w'''(l,t) = 0. \quad (17)$$

III. CONTROL FORMULATION

This section discusses the development of the control scheme for the suppression of the vibrations of the rod during its underwater transport. Several active control strategies are available for application to the RM system [18-32]. However, in this paper, a boundary control scheme (i.e., feedback control) is developed to suppress the vibrations (i.e., $u(x,t)$ and $w(x,t)$) of the rod under the effect of time-varying distributed hydrodynamic loads. Based on Lyapunov's direct method [14], the following control inputs to the bridge and the trolley $f_y(t)$ and $f_z(t)$, respectively, are designed such that the uniform ultimate boundedness of the closed loop system is achieved.

$$f_y(t) = -k_1 \dot{y}(t) - k_2 u'''(0,t)^2 - k_3 \dot{u}(l,t)^2 + k_4 u'(l,t)^2, \quad (18)$$

$$f_z(t) = -k_5 \dot{z}(t) - k_6 w'''(0,t), \quad (19)$$

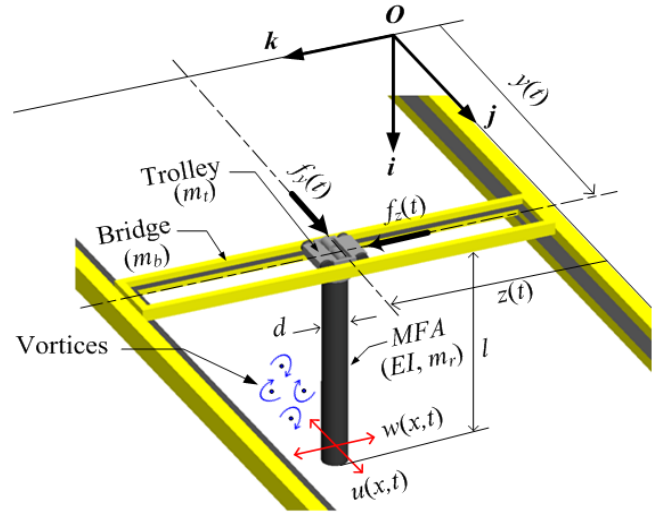


Fig. 1. Schematic of a nuclear refueling machine [18].

where $k_n (n=1,2,\dots,6)$ are positive gains. For notational convenience, instead of $\dot{w}(x,t)$ and $w'(x,t)$, $\dot{w}$ and w' are used, and similar abbreviations are employed subsequently.

In a recent publication [18], the control input $f_y(t)$ to the bridge required the feedback signal ($u'''(0,t)$), related to the shear force at the top end of the rod, for suppressing the lateral vibrations of the rod. However, application of such control resulted in large bending of the rod in the lateral direction. Therefore, in this paper, to reduce both the vibrations and bending of the rod, the feedback of the squared terms ($u'''(0,t)^2$, $\dot{u}(l,t)^2$, and $u'(l,t)^2$) is included in the control input $f_y(t)$ to the bridge.

A. Stability Analysis

Young's inequality along with the following lemma of bounded input and bounded output is utilized in analysing the stability of the system.

Lemma 1 [33]. Let $\varphi(x,t) \in R$ be a function defined on $x \in [0,l]$ and $t \in [0,\infty)$ that satisfies the boundary condition $\varphi(0,t) = 0$, for $\forall t \in [0,\infty)$, then the following inequalities hold.

$$\int_0^l \varphi^2 dx \leq l^2 \int_0^l \varphi'^2 dx, \varphi^2 \leq l \int_0^l \varphi'^2 dx, \forall x \in [0,l]. \quad (20)$$

If $\varphi(x,t)$ satisfies the boundary conditions $\varphi(0,t) = 0$ and $\varphi'(0,t) = 0$, for $\forall t \in [0,\infty)$, then the following inequalities also hold.

$$\int_0^l \varphi'^2 dx \leq l^2 \int_0^l \varphi''^2 dx, \varphi'^2 \leq l^3 \int_0^l \varphi''^2 dx, \forall x \in [0,l]. \quad (21)$$

Based on the total mechanical energy of the system, the Lyapunov candidate function

$$V(t) = V_0(t) + V_1(t) + V_2(t), \quad (22)$$

is introduced, where

$$V_0(t) = T + U = (1/2) \left(EI \int_0^l (u''^2 + w''^2) dx + (m_b + m_t) \dot{y}^2(t) + m \int_0^l (\dot{y}(t) + \dot{u})^2 dx + m_t \dot{z}^2(t) + m \int_0^l (\dot{z}(t) + \dot{w})^2 dx \right), \quad (23)$$

$$V_1(t) = \alpha_1 \int_0^l \dot{u}^2 dx + \alpha_2 \int_0^l \dot{w}^2 dx + \alpha_3 \int_0^l u''^2 dx + \alpha_4 \int_0^l w''^2 dx, \quad (24)$$

$$V_2(t) = m\beta \int_0^l (u\dot{u} + w\dot{w}) dx + \beta_1 \int_0^l u(u - \dot{y}(t)) dx + \beta_2 \int_0^l u(u + \dot{y}(t)) dx + \beta_3 \int_0^l w(w - \dot{z}(t)) dx + \beta_4 \int_0^l w(w + \dot{z}(t)) dx + \beta_5 \int_0^l \dot{y}(t)(\dot{y}(t) + \dot{u}) dx + \beta_6 \int_0^l \dot{z}(t)(\dot{z}(t) + \dot{w}) dx. \quad (25)$$

Lemma 2. The Lyapunov candidate function can be upper and lower bounded as

$$0 \leq \gamma_1 W(t) \leq V(t) \leq \gamma_2 W(t), \quad (26)$$

where

$$W(t) = \dot{y}^2(t) + \dot{z}^2(t) + \int_0^l (\dot{u}^2 + \dot{w}^2 + u''^2 + w''^2) dx, \quad (27)$$

and γ_1 and γ_2 are positive constants.

Proof. The proof of a similar lemma considering only the lateral dynamics (15) and control law (18) was given in [19]. Therefore, the proof of Lemma 2 is not included in this paper.

Lemma 3. The time derivative of the Lyapunov function in (22) can be upper bounded as follows.

$$\dot{V}(t) \leq -\gamma V(t) + \varepsilon, \quad (28)$$

where γ and ε are positive constants.

Proof. The proof of a similar lemma considering only the lateral dynamics (15) and control law (18) was given in [19]. Therefore, the proof of Lemma 3 is also not included in this paper.

Lemma 4 [14]. Let $V(t) \in R$ be a non-negative function of time on $t \in [0, \infty)$ that satisfies the differential inequality $\dot{V}(t) \leq -\gamma V(t) + \varepsilon$, where γ and ε are positive constants: Therefore,

$$V(t) \leq V(0)e^{-\gamma t} + \frac{\varepsilon}{\gamma}(1 - e^{-\gamma t}), \quad \forall t \in [0, \infty). \quad (29)$$

Theorem 1. For the system given by (15)-(16) and boundary conditions (17), under Assumptions 1-3 and control laws (18)-(19), given that the initial conditions are bounded and that the required state information (i.e., $y(t)$, $\dot{y}(t)$, $z(t)$, $\dot{z}(t)$, $u'''(0, t)$ and $w'''(0, t)$) is available, we can conclude the closed-loop system is uniform ultimate bounded and that all the close-loop signals are bounded.

Proof. Considering equation (22), Lemma 1, and Lemma 2, it is evident that

$$((EI + 2\alpha_3)/2l^3)u^2 \leq ((EI + 2\alpha_3)/2) \int_0^l u''^2 dx \leq V(t), \quad (30)$$

$$((EI + 2\alpha_4)/2l^3)w^2 \leq ((EI + 2\alpha_4)/2) \int_0^l w''^2 dx \leq V(t), \quad (31)$$

The inequalities above and Lemma 4 reveal that, $\forall x \in [0, L]$, the rod vibrations are uniformly bounded as follows.

$$|u(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_3))(V(0)e^{-\gamma t} + (\varepsilon/\gamma))}, \quad (32)$$

$$|w(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_4))(V(0)e^{-\gamma t} + (\varepsilon/\gamma))}. \quad (33)$$

Furthermore, from (32)-(33), we obtain

$$\lim_{t \rightarrow \infty} |u(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_3))(\varepsilon/\gamma)}, \quad (34)$$

$$\lim_{t \rightarrow \infty} |w(x, t)| \leq \sqrt{(2l^3/(EI + 2\alpha_4))(\varepsilon/\gamma)}. \quad (35)$$

From (34)-(35), it can be seen that both $u(x, t)$ and $w(x, t)$ will converge exponentially to a small neighbourhood of zero if the values of parameters γ , α_3 , and α_4 are large. Therefore, there exists a time T_0 , such that for $\forall t \geq T_0$, the boundary output converges to a small neighbourhood around zero. According to Assumptions 1-2, $V_0(t)$ (i.e., the total mechanical energy of the RM) is bounded, therefore, all the signals $\dot{y}(t)$, $\dot{z}(t)$, u'' , w'' , u''' and w''' are bounded. Hence, it is concluded that the control laws (18)-(19) are also bounded. $\square$

IV. SIMULATION

This section presents the simulations of the responses of the rod in water caused by the movements of the RM. An implicit finite difference scheme was developed to solve the equations of motion, (15)-(16), numerically. Then, the responses $u(x, t)$ and $w(x, t)$ of the rod upon the movements of the bridge and the trolley were obtained in MATLAB. The parameters given in Table I were used for simulations.

First, the responses of the rod were obtained when the bridge was moved with a trapezoidal velocity command shown in Fig. 2b. Fig. 2a depicts the acceleration of the bridge. The lateral $u(l, t)$ and transverse $w(l, t)$ endpoint deflections of the rod upon the trapezoidal command are given in Figs. 2c and 2d, respectively. It can be seen that the rod exhibits significant residual and transverse vibrations when the bride is moved with the trapezoidal command. Now, to demonstrate the vibration suppression capability of the proposed control scheme, (18)-(19), the lateral and transverse endpoint deflections of the rod are represented by the blue dash-dot lines in Figs. 3c and 3d, respectively. Where the control inputs to the bridge $f_y(t)$ and the trolley $f_z(t)$ are shown in Figs. 3a and 3b, respectively. The green dashed line in Fig. 3c shows the lateral endpoint deflection of the rod

TABLE I
SIMULATION PARAMETERS [18]

Parameters	Units	Values
Rod length (l)	m	1.0
Rod diameter (d)	m	0.008
Rod mass (m_r)	kg	0.037
Trolley mass (m_t)	kg	3.0
Bridge mass (m_b)	Kg	7.0
Young's modulus (E)	GPa	3.89
Density of water (ρ_w)	kg/m ³	1000
Density of fuel rod (ρ_r)	kg/m ³	1204.5
Added-mass coefficient (C_a) [20]	-	0.93
Drag coefficient (C_d) [20]	-	1.28

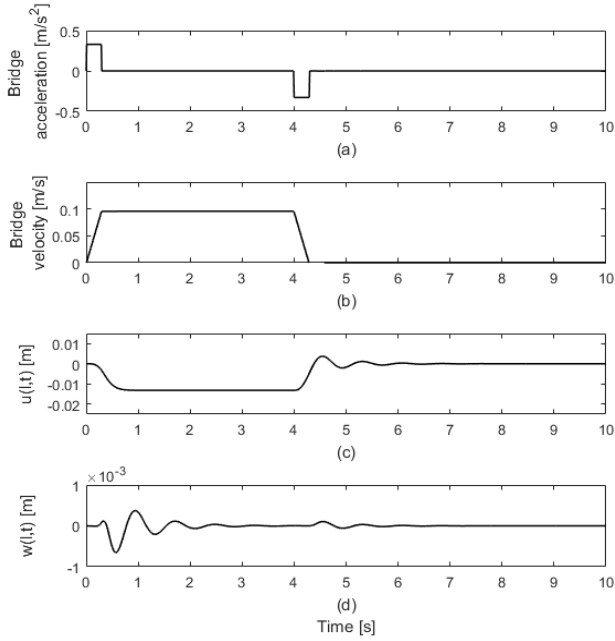


Fig. 2 Underwater responses of fuel rod to a trapezoidal command (without control): (a) Bridge acceleration; (b) bridge velocity; (c) lateral endpoint deflections of rod; (d) transverse endpoint deflections of rod.

when the control input from [18] was used to drive the bridge of the RM. The 3D lateral and transverse responses of the rod to the trapezoidal velocity command, shown in Fig. 2b, are presented in Fig. 4. Whereas those to the proposed boundary control scheme are shown in Fig. 5. It can be seen that the proposed control scheme (18)-(19) can successfully suppress the transient and residual vibrations of the rod.

V. RESULTS AND CONCLUSIONS

This paper presented the vibration control problem of the nuclear RM, which transports nuclear fuel rods to their UW target positions within the nuclear reactor. First, a hybrid RM model consisting of two lumped masses (i.e., the bridge the trolley) and a flexible MFA system was developed in

consideration of the distributed nonlinear hydrodynamic force acting on the MFA. Hamilton's principle was utilized in obtaining the equations of motion of the considered system. Furthermore, simulations were performed to reveal that the rod exhibited significant residual vibrations as a trapezoidal command was used to transport the bridge (see the responses in Figs. 2c, 2d, 4a, and 4b). Then, a boundary feedback control scheme was developed given by equations (18)-(19), which suppressed both the lateral and transverse vibrations of the rod, significantly (see the responses in Figs. 3c, 3d, 5a, and 5b).

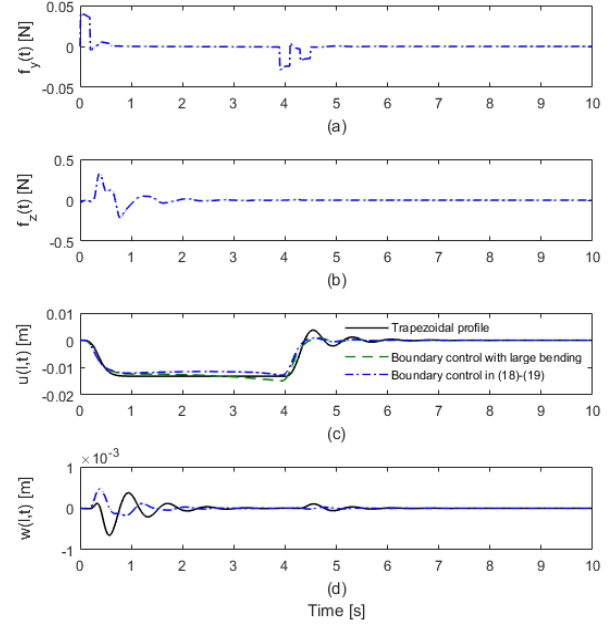


Fig. 3 Underwater responses of fuel rod with boundary control law: (a) Control input to bridge; (b) control input to trolley; (c) lateral endpoint deflections of rod; (d) transverse endpoint deflections of rod.

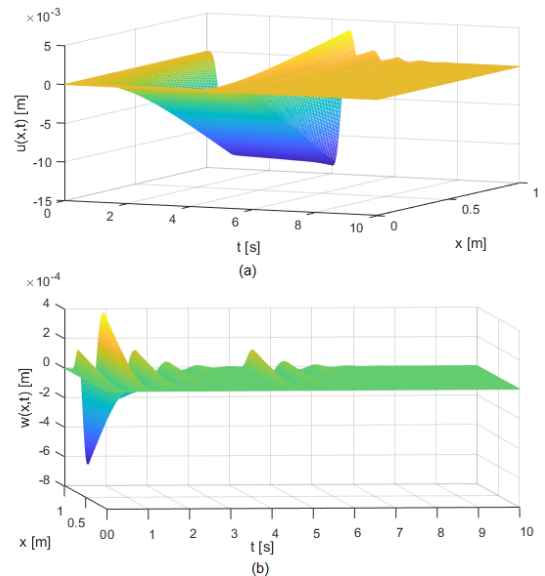


Fig. 4 3D responses of the rod to the trapezoidal command (without control): (a) Lateral deflections; (b) transverse deflections.

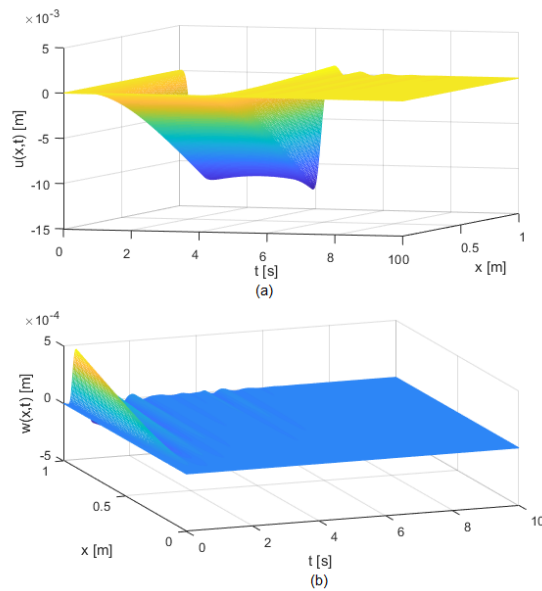


Fig. 5 3D responses of the rod with boundary control: (a) Lateral deflections; (b) transverse deflections.

In conclusion, a hybrid rigid-flexible model of a nuclear RM was developed based on Hamilton's principle for investigation of the fuel-rod responses in water generated by RM movements. Further, a boundary control law was developed to suppress rod vibrations. A Lyapunov-based stability analysis of the closed-loop system was performed to show that the rod vibrations are uniformly ultimately bounded under the proposed control law in the presence of external disturbances (i.e., nonlinear hydrodynamic forces). Finally, numerical simulations, provided in this paper, were run to verify the effectiveness of the proposed boundary control scheme.

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